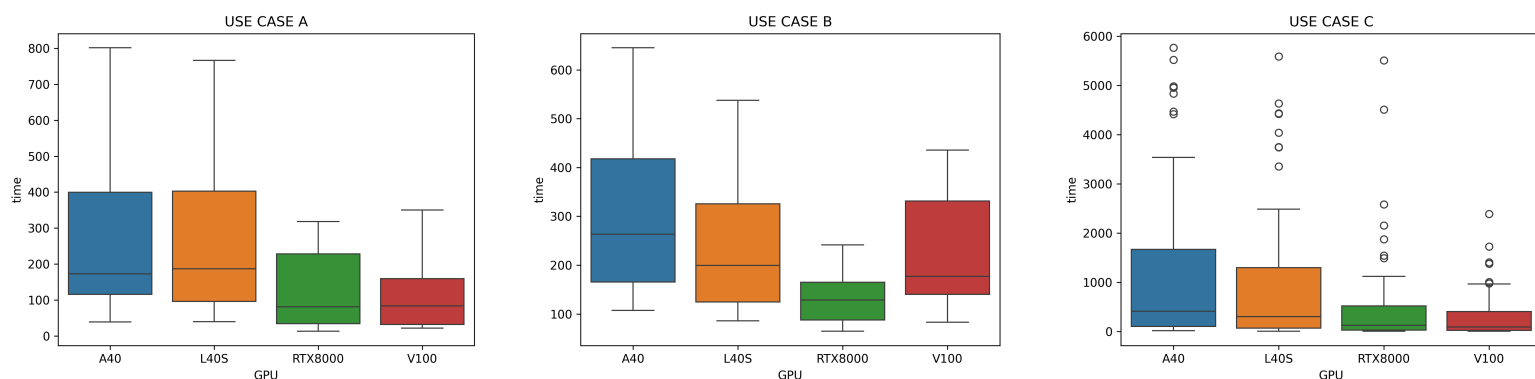


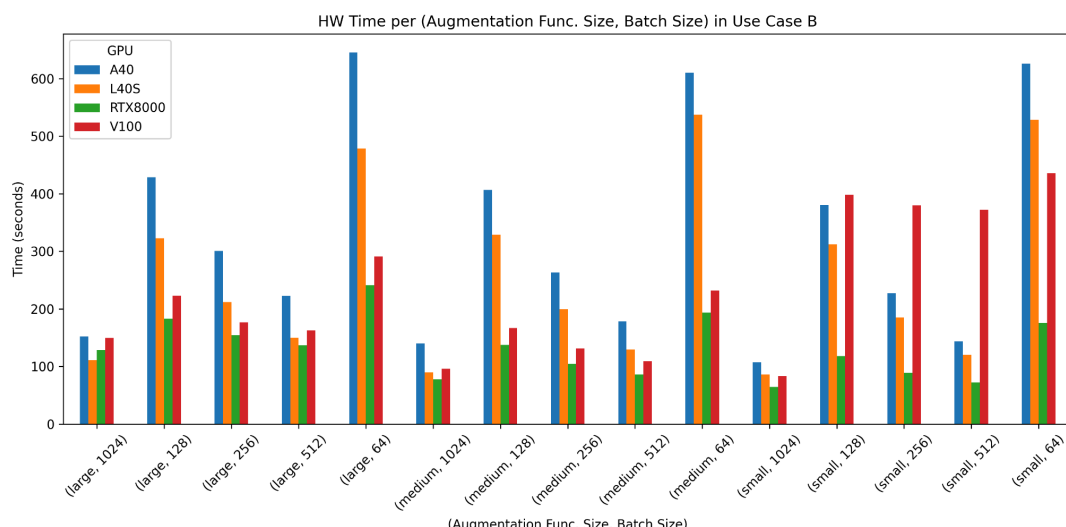
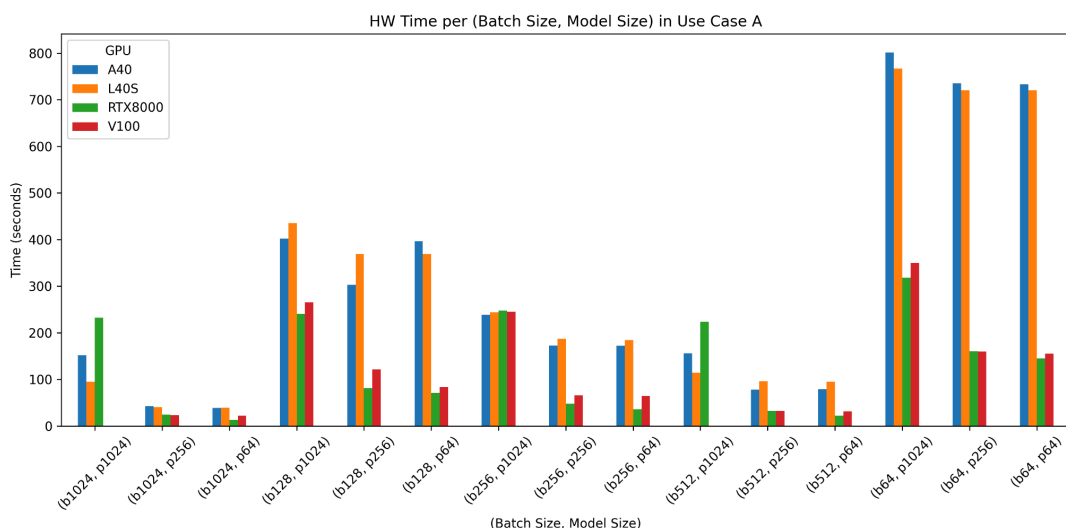
FBK Analysis

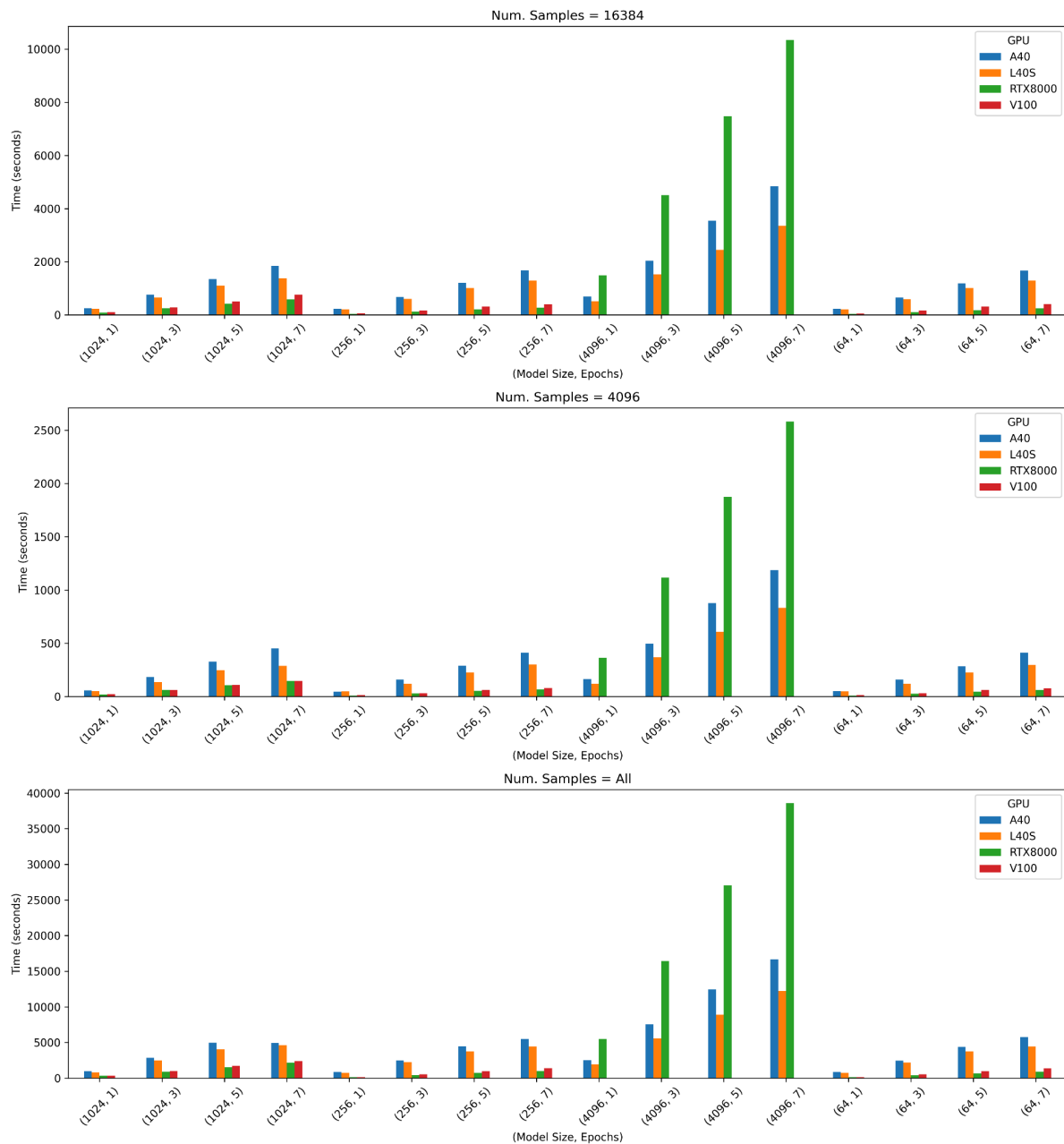
I noticed I don't seem to have the cliffjumper results (or at least I can't find any GPU with that codename), so I guessed the difference mentioned was between the A40 and RTX or V100. My first hypothesis was that small experiments create high variability, which is also compounded (probably) by the fact that nodes are not exclusive on this cluster.



Overall tendency with a high number of experiments seems correct, but there is large variability (and outliers) in Use Case C (UC_C). I also removed ~15 outliers from UC_A and some from UC_C for visualization purposes.

Starting from this, I decided to look at execution times after pairing runs based on equivalent variables (all runs for all HWs where model params=1024 and batch size=128). I get:



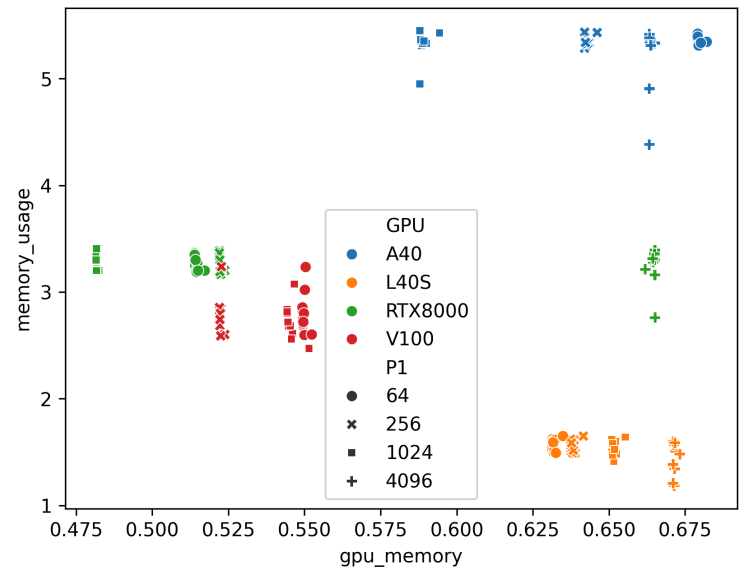
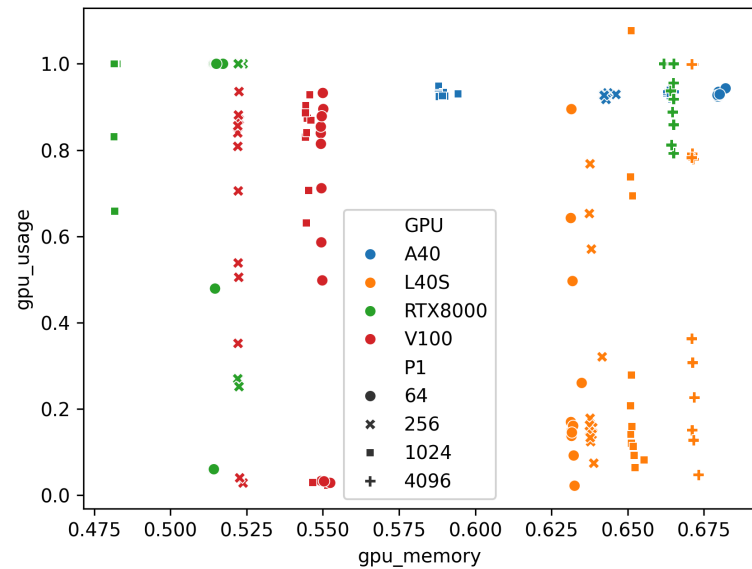


What I noticed:

- The tendency in all use cases seems to indicate to me that RTX and V100 are performing better *in general* compared to A40 and L40S;
- One exception to this is in UC_C with large model size = 4096, where *I guess* the RTX RAM gets filled and V100 goes OOM completely. (I'll check this later with gpu stats **[1]**);
- Another exception is small batch size seems to slow down L40S and A40 (or be handled much better by the other two GPUs), this can be seen both in UC_A and in UC_B **[2]**;
- The last exception is in UC_B, where the V100 seems to handle large computation well and to slow down with frequent open() close() operations **[3]**.

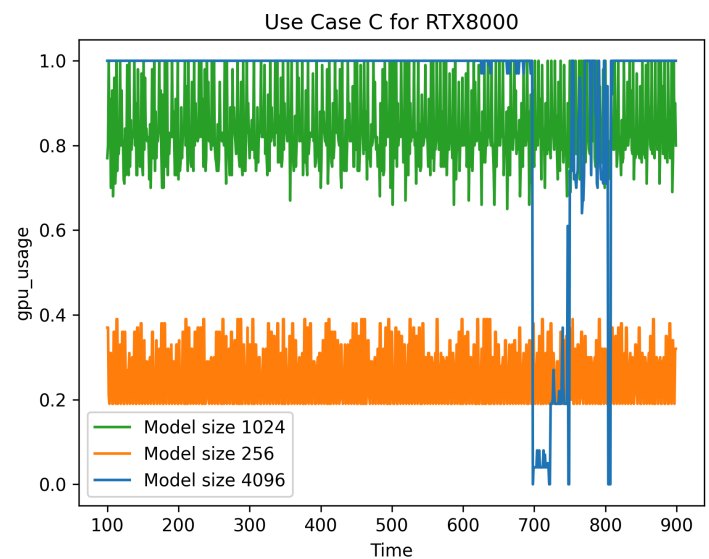
[1] RTX RAM with large model size

RAM with model size = 4096 has more RAM usage than all others?

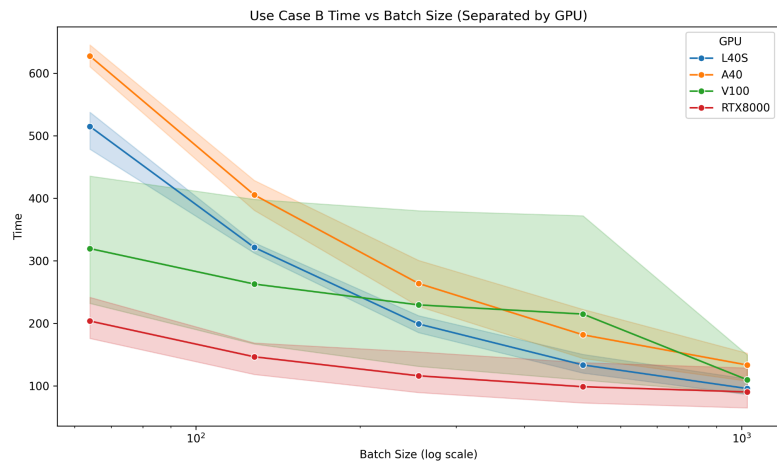
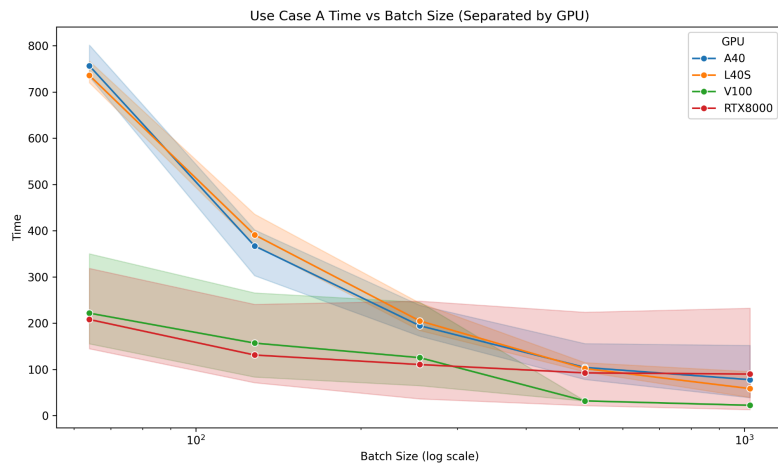


Doesn't seem the case to me, but could be because the gpu_usage is at 100% in that case.

Time series seems to confirm that as well.



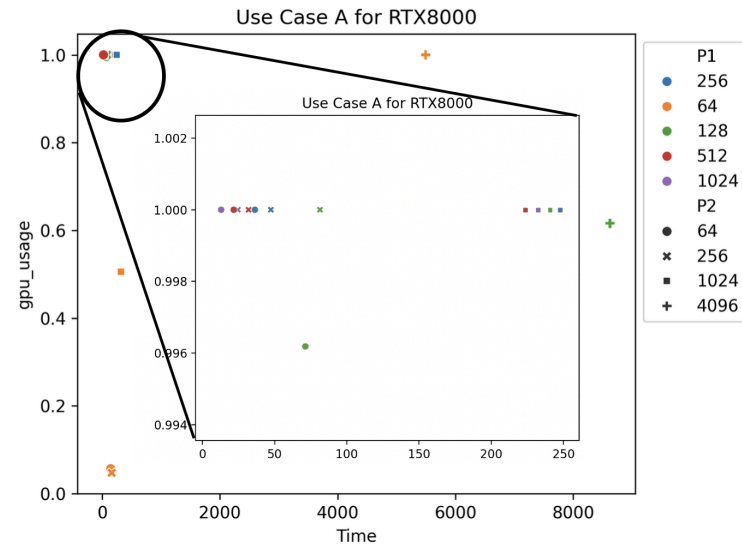
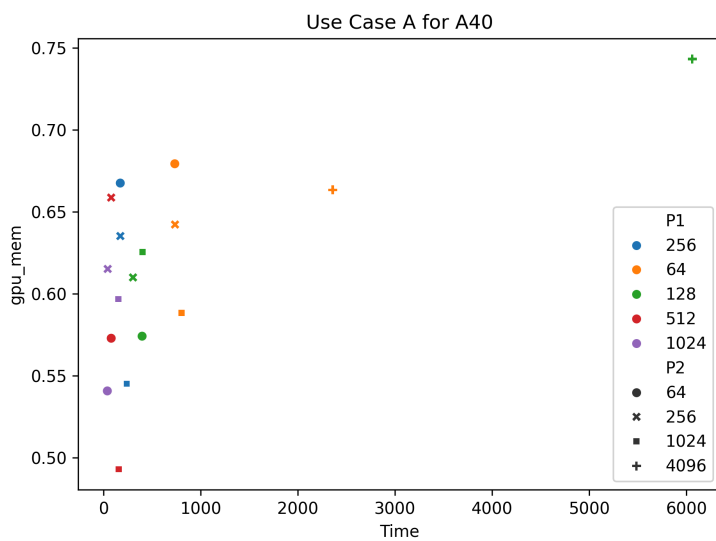
[2] Small batch sizes with L40S and A40



I started plotting the correlation between time and batch size. It seems the V100 and RTX GPUs handle data better than the other two until batch size=1024, then performance gets closer.

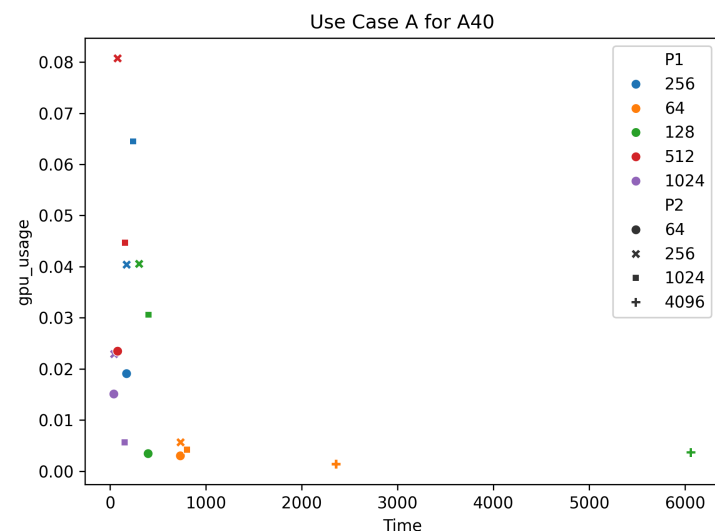
I'll check if here also we reached the max compute or max mem, it seems it's not L40S and A40 being slower but the other two scaling worse with more data.

Also, I don't think that it's memory which is influencing the execution time, otherwise there would be more correlation (chart below on the left).



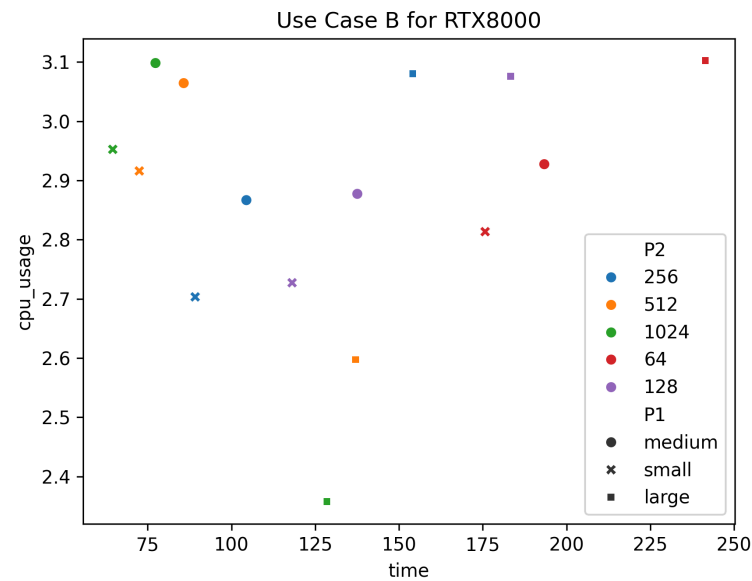
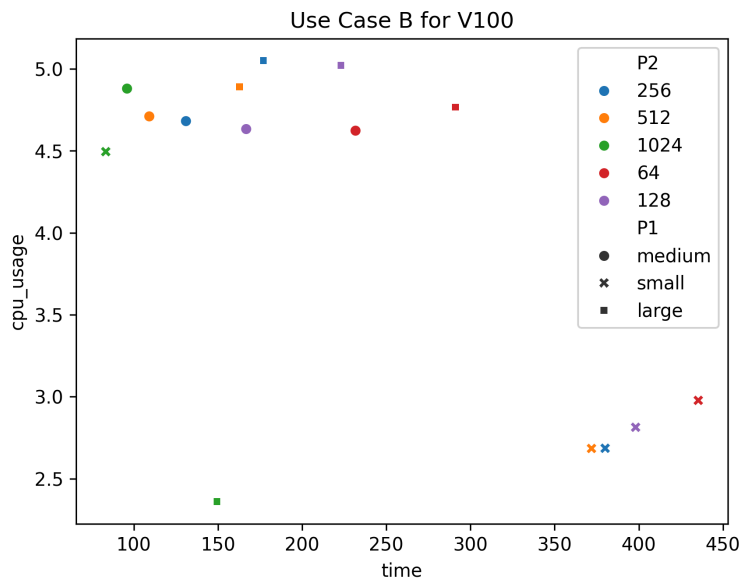
On the other hand the gpu usage is at 100% for almost all the runs in UC_A (for RTX8000), which is also strange, maybe it reached the memory bound?

If we inspect one of the better scaling GPUs (A40 in this case) the gpu_usage is very low for all runs, which I think explains why this one scales better.



[3] V100 and I/O

This one is a bit more strange, I can't really think of anything except I/O impact.



Plotting the performance however it seems this is an issue isolated to V100 with heavy I/O operations. For example, checking the RTX, this one has a much smoother distribution of performance compared to the V100, which has only small augmentation runs which take a long time and all others which are faster and use more cpu.

The same behaviour (more smooth distribution) persists even with the other GPUs